Expt. No. 10

Title: To design the layout of Carry look ahead adder using DSCH and Micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

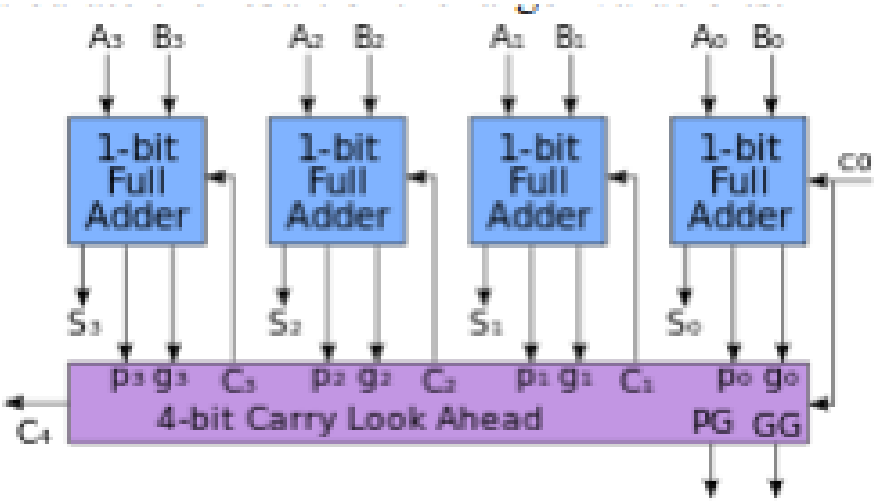
Experiment No. 10

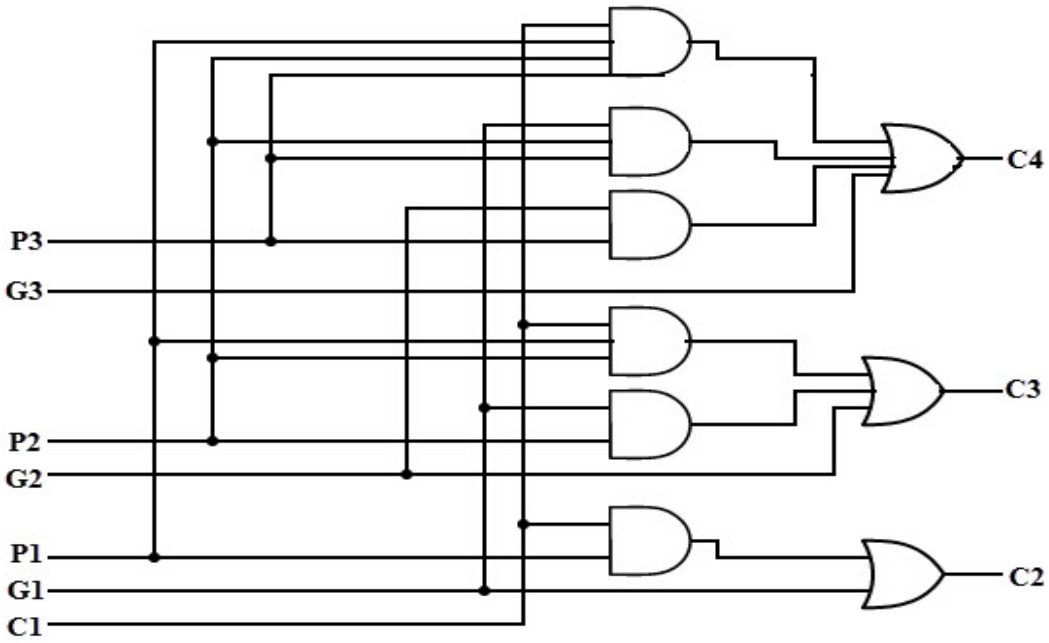
Aim: To design the layout of Carry look ahead adder using DSCH and Micro-wind

Software Required: DSCH and Micro-wind software

Hardware: computer

Circuit Diagram:



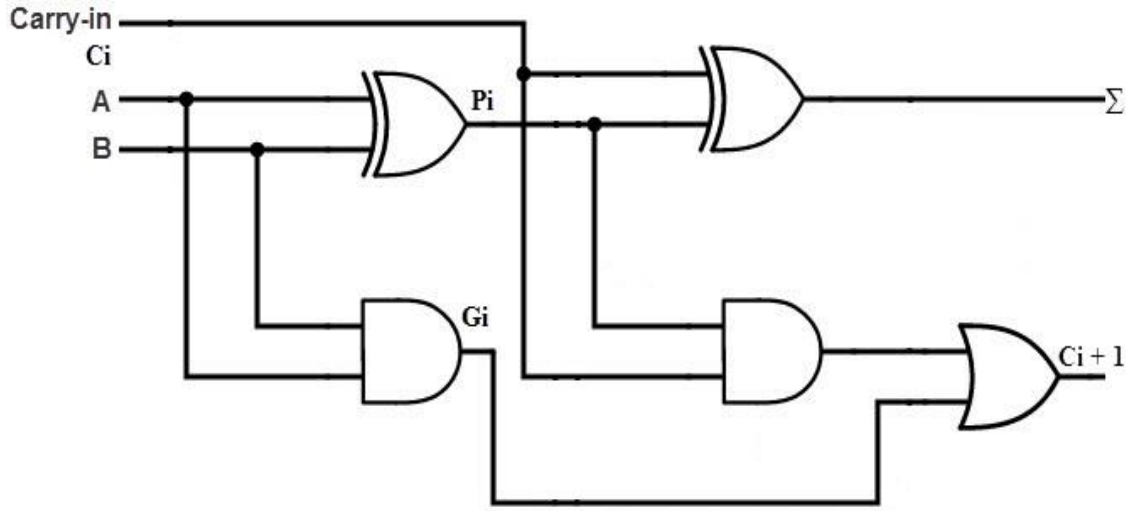


Theory:

Carry-Look ahead Adder

A carry-Lookahead adder is a fast parallel adder as it reduces the propagation delay by more complex hardware, hence it is costlier. In this design, the carry logic over fixed groups of bits of the adder is reduced to two-level logic, which is nothing but a transformation of the ripple carry design.

This method makes use of logic gates so as to look at the lower order bits of the augend and addend to see whether a higher order carry is to be generated or not. Let us discuss in detail.



A	B	Ci	C i +1	Condition
0	0	0	0	No carry generate
0	0	1	0	
0	1	0	0	
0	1	1	1	No carry propagate
1	0	0	0	
1	0	1	1	
1	1	0	1	Carry generate
1	1	1	1	

Consider the full adder circuit shown above with corresponding truth table. If we define two variables as carry generate Gi and carry propagate Pi then,

$P_i = A_i \oplus B_i$

Gi = Ai Bi

The sum output and carry output can be expressed as

$S_i = P_i \oplus C_i$

$C_{i+1} = G_i + P_i C_i$

Where Gi is a carry generate which produces the carry when both Ai, Bi are one regardless of the input carry. Pi is a carry propagate and it is associate with the propagation of carry from Ci to Ci +1.

The carry output Boolean function of each stage in a 4 stage carry-Lookahead adder can be expressed as

$C1 = G0 + P0\ Cin$

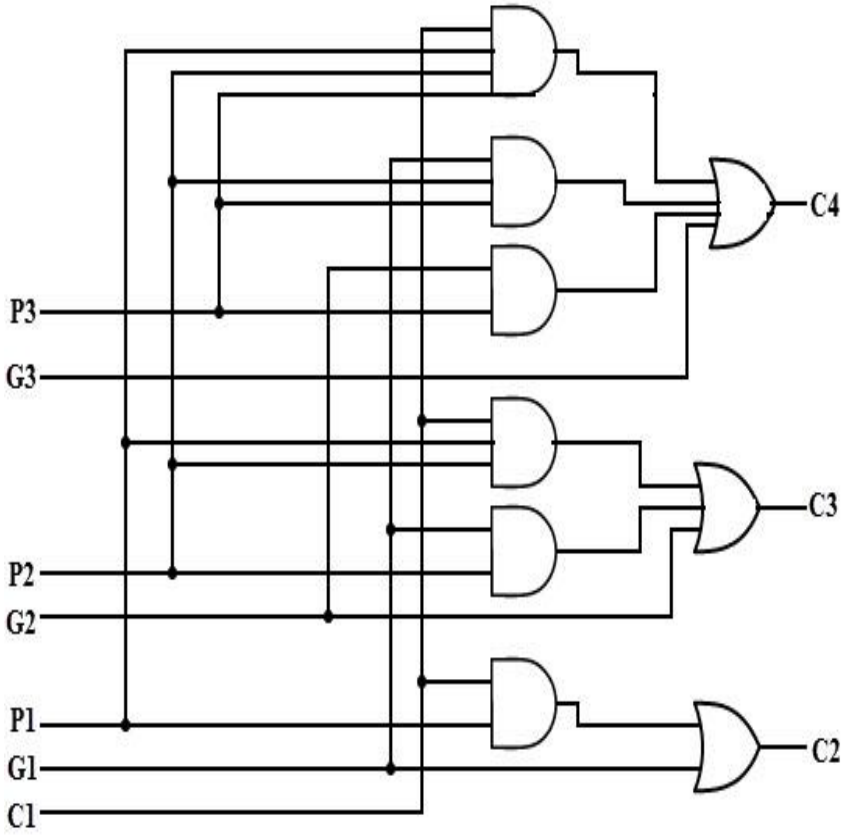
$C2 = G1 + P1\ C1 = G1 + P1\ G0 + P1\ P0\ Cin$

$C3 = G2 + P2\ C2 = G2 + P2\ G1 + P2\ P1\ G0 + P2\ P1\ P0\ Cin$

$C4 = G3 + P3\ C3 = G3 + P3\ G2 + P3\ P2\ G1 + P3\ P2\ P1\ G0 + P3\ P2\ P1\ P0\ Cin$

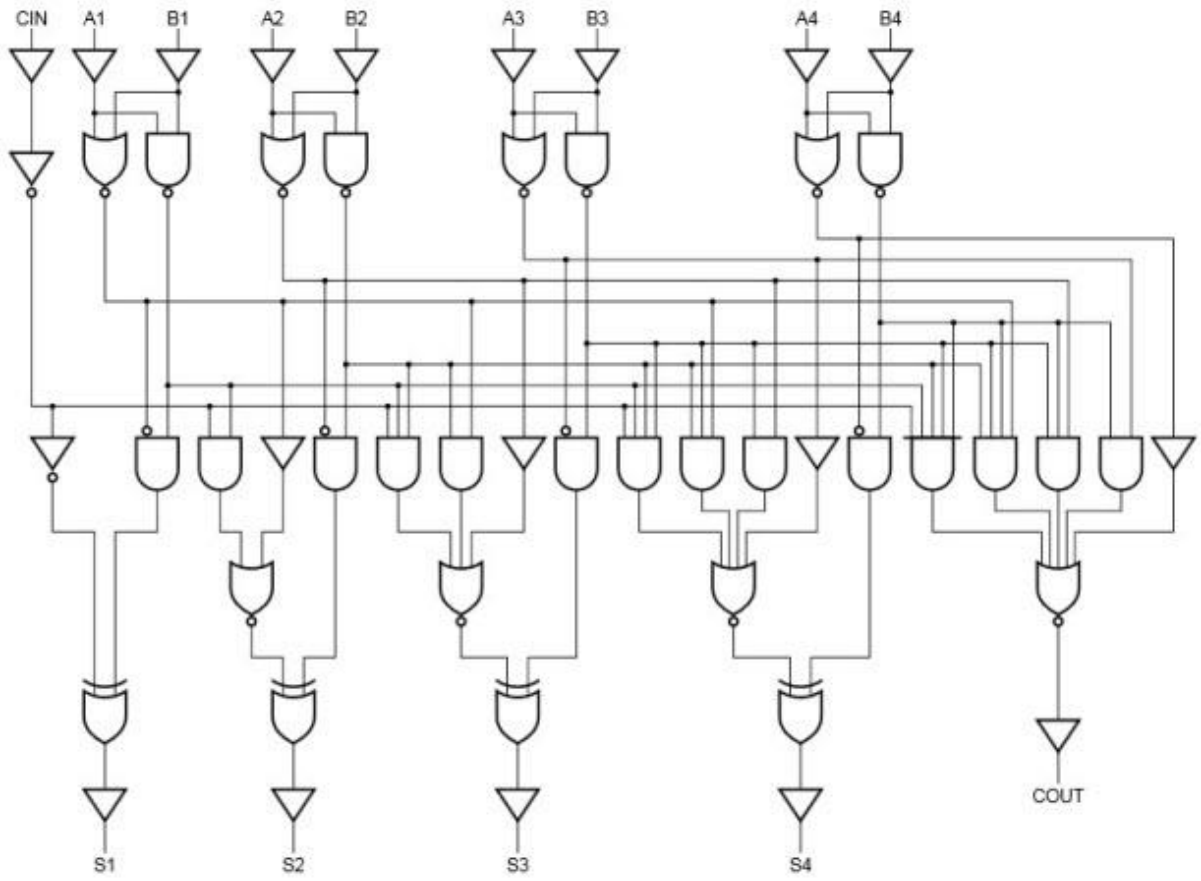
From the above Boolean equations, we can observe that C4 does not have to wait for C3 and C2 to propagate but actually C4 is propagated at the same time as C3 and C2. Since the Boolean expression for each carry output is the sum of products so these can be implemented with one level of AND gates followed by an OR gate.

The implementation of three Boolean functions for each carry output (C2, C3 and C4) for a carry-Lookahead carry generator shown in below figure.



Therefore, a 4 bit parallel adder can be implemented with the carry-Lookahead scheme to increase the speed of binary addition as shown in below figure. In this, two Ex-OR gates are required by each sum output. The first Ex-OR gate generates Pi variable output and the AND gate generates Gi variable.

Hence, in two gates levels all these P's and G's are generated. The carry-Lookahead generators allows all these P and G signals to propagate after they settle into their steady state values and produces the output carriers at a delay of two levels of gates. Therefore, the sum outputs S2 to S4 have equal propagation delay times.



It is also possible to construct 16 bit and 32 bit parallel adders by cascading the number of 4 bit adders with carry logic. A 16 bit carry-Lookahead adder is constructed by cascading the four 4 bit adders with two more gate delays, whereas the 32 bit carry-Lookahead adder is formed by cascading of two 16 bit adders.

In a 16 bit carry-Lookahead adder, 5 and 8 gate delays are required to get C16 and S15 respectively, which are less as compared to the 9 and 10 gate delay for C16 and S15 respectively in cascaded four bit carry-Lookahead adder blocks. Similarly, in 32 bit adder, 7 and 10 gate delays are required by C32 and S31 which are less compared to 18 and 17 gate delays for the same outputs if the 32 bit adder is implemented by eight 4 bit adders.

Carry-Look ahead Adder ICs

The high-speed carry-Lookahead adders are integrated on integrated circuits in different bit configurations by several manufacturers. There are several individual carry generator ICs are available so that we have to make connection with logic gates to perform the addition operation.

A typical carry-Lookahead generator IC is 74182 which accept four pairs of active low carry propagate (as P0, P1, P2 and P3) and carry generate (Go, G1, G2 and G3) signals and an active high input (Cn).

It provides active high carriers (Cn+x, Cn+y, Cn+z) across the four groups of binary adders. This IC also facilitates the other levels of look ahead by active low propagate and carry generate outputs.

The logic expressions provided by the IC 74182 are

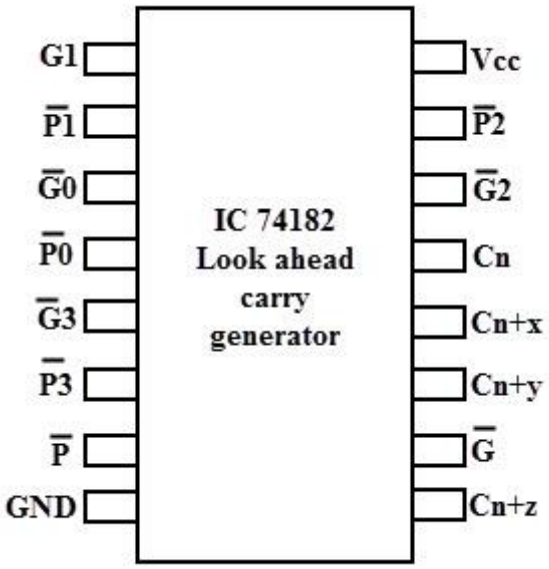
$$C_{n+x} = G_0 + P_0 C_n$$

$$C_{n+y} = G_1 + P_1 G_0 + P_1 P_0 C_n$$

$$C_{n+z} = G_2 + P_2 G_1 + P_2 P_1 G_0$$

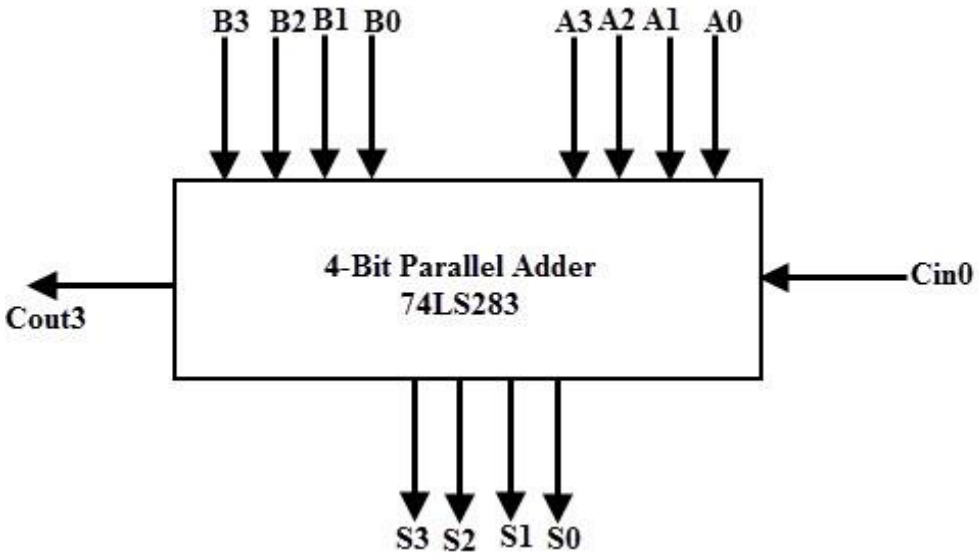
$$\overline{G} = \overline{G3 + P3\,G2 + P3\,P2\,G1 + P3\,P2\,P1\,G0}$$

$$\overline{P} = \overline{P3\,P2\,P1\,P0}$$



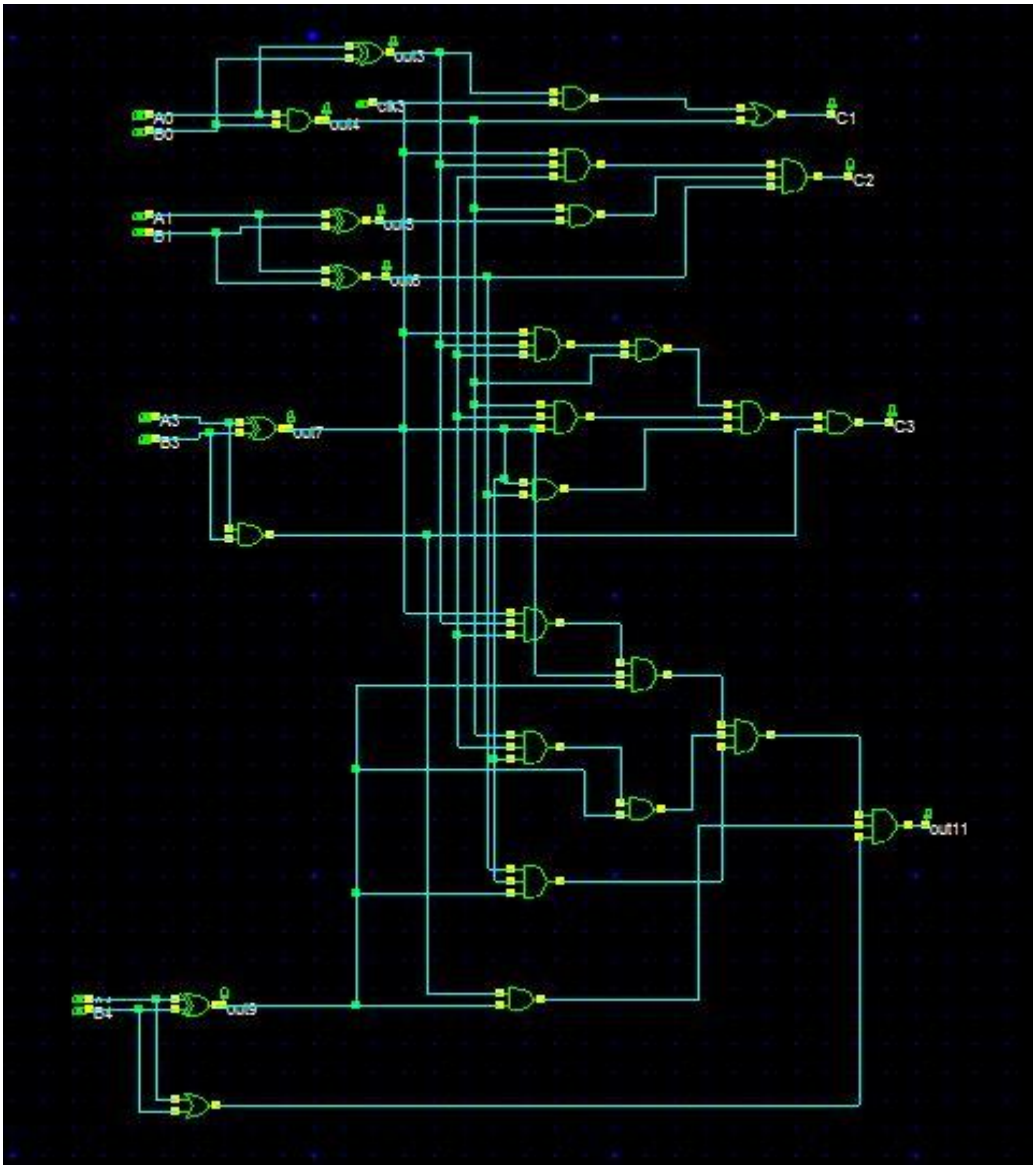
On the other hand, there are many high speed adder ICs which combine a set of full adders with carry-Lookahead circuitry. The most popular form of such IC is 74LS83/74S283 which is a 4 bit parallel adder high speed IC that contains four interconnected full adders with a carry-Lookahead circuitry.

The functional symbol for this type of IC is shown in below figure. It accepts the two 4 bit numbers as A3A2A1A0 and B3B2B1B0 and input carry Cin0 into the LSB position. This IC produce output sum bits as S3S2S1S0 and the carry output Cout3 into the MSB position.



y cascading two or more parallel adder ICs we can perform the addition of larger binary numbers such as 8 bit, 24-bit and 32 bit addition.

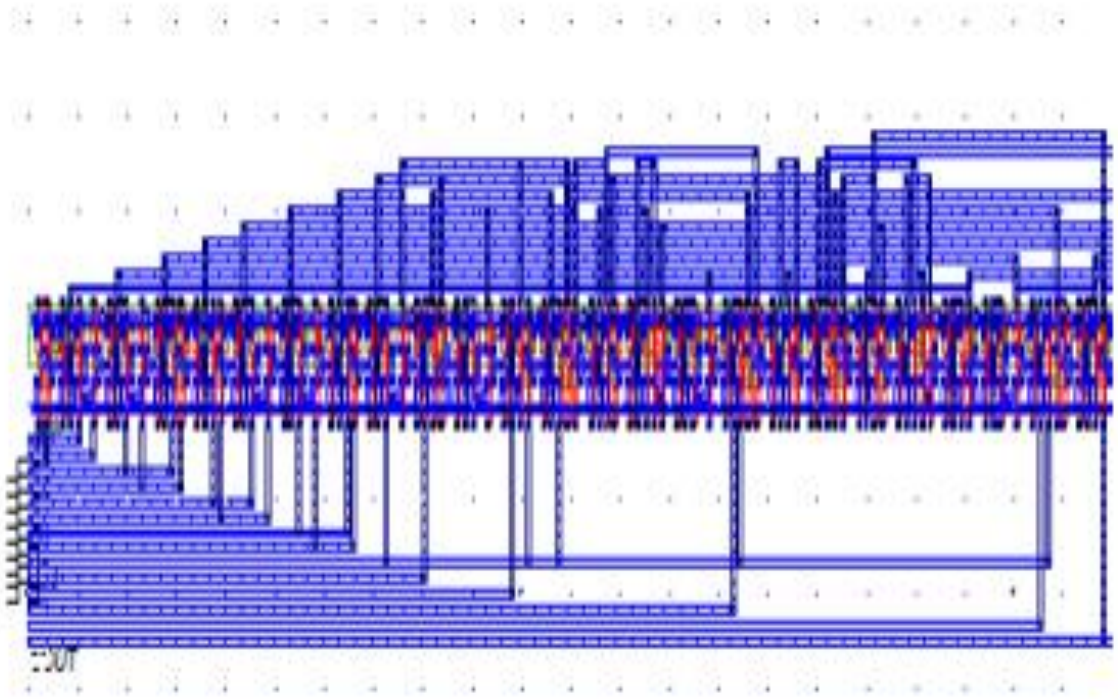
Diagram:



Procedure:

1. Open Micro-wind DSCH Software.
2. Draw the Ripple Carry Adder following the design
3. Save the circuit and make verilog file.
4. Open micro-wind and compile the verilog file.
5. Observe layout and run simulation.

Layout:



Pre-Lab Questions:

1 What distinguishes the look-ahead-carry adder?

Post Lab Questions:

1. Carry look-ahead logic uses the concepts of _____

Conclusion: Four bit Carry look ahead adder has been successfully implemented on DSCH and Code has been tested in micro-wind and layout recorded.

CO's covered:

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